

HOSPITAL OPERATIONS & PATIENT ANALYTICS DASHBOARD

DASHBOARD TESTING AND VALIDATION REPORT

Submitted as part of the Infosys Springboard Internship Project

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1. Introduction

The Hospital Operations & Patient Analytics Dashboard was built to give hospital administrators a single, centralized view of operations through a suite of interactive dashboards and healthcare performance indicators. Before it was signed off for final submission, the dashboard went through a thorough round of testing to confirm that it worked correctly, reported accurate figures, and was genuinely usable.

The goal of testing was straightforward: make sure every dashboard component behaved as intended, no matter how a user interacted with it. Each dashboard was reviewed to confirm that KPI values were calculated correctly, that visualizations faithfully represented the underlying data, and that interactive elements such as slicers and filters responded consistently. Particular care was taken with healthcare-specific metrics, departmental comparisons, patient flow analysis, and resource utilization, since these are the numbers that hospital teams would actually rely on.

Testing covered all four dashboards developed during the project — the Executive Hospital Performance Dashboard, Patient Flow Analytics, Department Analytics, and Resource Utilization. Every KPI, chart, filter, and interactive element was checked against the cleaned hospital dataset to confirm that the information displayed stayed accurate and consistent across different filter selections.

Testing began once dashboard development and integration were complete. A series of validation scenarios were run to confirm data accuracy, responsiveness, visual consistency, and interaction behaviour, and any minor formatting or visualization issues found along the way were fixed before the dashboard was finalized.

This report walks through the testing approach, the validation procedures used, and the outcomes observed. It demonstrates that the dashboard suite meets the functional requirements set for the project and is ready for analytical reporting, operational monitoring, and decision-making within the scope of this healthcare analytics solution.

2. Testing Objectives

The testing process set out to verify that the Hospital Operations & Patient Analytics Dashboard performs accurately, consistently, and reliably under a range of user interactions — validating both the analytical logic behind the dashboard and the visual components that present it.

- Verify the accuracy of every healthcare KPI, including Total Admissions, Total Patients, Total Revenue, Average Length of Stay (LOS), Readmission Rate, Bed Utilization Rate, Occupancy Rate, and Department Efficiency Score.
- Confirm that charts, graphs, and visualizations accurately represent the processed hospital dataset without inconsistencies or incorrect aggregations.
- Validate interactive elements such as date range slicers, department filters, gender filters, ward filters, and equipment filters, along with other report-level interactions.

- Confirm that interactions update the relevant visuals dynamically while keeping data consistent across all applicable reports.
- Evaluate the correctness of DAX measures and calculated fields by comparing displayed values against the processed healthcare dataset.
- Verify that all four dashboards operate independently while staying consistent when users apply common filters across views.
- Assess usability — proper alignment of visual elements, readable labels, intuitive navigation, and clear presentation of analytical information.
- Identify any functional, visual, or performance issues before final submission and confirm the dashboard suite is ready for operational use.
- Confirm the completed dashboard meets the testing and validation requirements defined for the project and delivers reliable insight for healthcare operations analysis.

3. Testing Scope

The scope of testing was defined to ensure every major component of the dashboard suite was thoroughly validated ahead of final submission — covering functional behaviour, data accuracy, user interaction, and overall performance across all four dashboards:

- **Executive Hospital Performance Dashboard**
- **Patient Flow Analytics**
- **Department Analytics**
- **Resource Utilization**

3.1 Dashboard Components

All visual components developed as part of the dashboard suite were verified, including:

- KPI Cards
- Bar Charts
- Column Charts
- Line Charts
- Donut Charts
- Treemap
- Interactive Slicers
- Data Labels
- Chart Titles

- Legends and Axes

3.2 KPI Validation

All calculated healthcare KPIs were tested to confirm the displayed values matched the processed dataset and corresponding DAX calculations, covering:

- Total Patients
- Total Admissions
- Total Revenue
- Average Length of Stay (LOS)
- Readmission Rate
- Total Readmissions
- Bed Utilization Rate
- Occupancy Rate
- Department Efficiency Score
- Equipment Availability
- Average Equipment Usage

3.3 Dashboard Functionality

Functional testing confirmed that dashboard features operated correctly, including:

- KPI calculations
- Chart rendering
- Department-wise analysis
- Monthly trend analysis
- Patient flow analysis
- Resource utilization analysis
- Department comparison
- Equipment analysis
- Staff allocation analysis

3.4 Interactive Features

Interactive functionality was verified to ensure a smooth user experience, including:

- Date range slicers
- Department filters
- Gender filters

- Equipment type filters
- Ward filters
- Equipment name filters
- Cross-filtering between visuals
- Dynamic KPI updates
- Visual refresh after filter selection

3.5 Data Validation

Data consistency checks confirmed that:

- Visualizations displayed accurate information from the processed dataset.
- Aggregated values matched the underlying data.
- Department-wise calculations remained consistent after applying filters.
- Monthly trend charts reflected the correct chronological sequence.
- KPI values updated correctly based on the selected filter context.

3.6 Performance Validation

Overall dashboard performance was evaluated to ensure:

- Dashboards loaded successfully without errors.
- Visuals rendered correctly.
- Filters responded within an acceptable time.
- Dashboard navigation remained smooth.
- Report layout remained consistent across all pages.

Together, these activities confirmed that the dashboard suite met the functional and validation requirements defined for the project while delivering accurate, reliable, interactive healthcare analytics — directly supporting the project's Module 7 focus on KPI validation, healthcare metrics verification, dashboard interaction testing, and patient flow validation.

4. Testing Environment

Testing was conducted in the same development environment used throughout the build, once data preparation, KPI development, dashboard integration, and report validation were complete.

The hospital dataset used during testing was cleaned and transformed through Python-based preprocessing, then stored in a structured Excel workbook and imported into Power BI, where table relationships were

established to support interactive reporting and analytical calculations. All KPI measures used during testing were built with DAX and validated against the processed dataset.

Component	Technology / Environment
Dashboard Development Tool	Microsoft Power BI Desktop
Data Processing	Python (Pandas, NumPy)
Data Source	Processed Hospital Dataset (.xlsx)
KPI Development	DAX (Data Analysis Expressions)
Programming Language	Python 3
Version Control	Git & GitHub
Operating System	Microsoft Windows
Dataset Type	Multi-table Hospital Operations Dataset
Dashboard Pages Tested	Executive Hospital Performance, Patient Flow Analytics, Department Analytics, Resource Utilization

Every filtering scenario applied during testing used the finalized dataset, confirming that KPIs, visualizations, filters, and analytical reports produced consistent, accurate results without affecting dashboard performance. Keeping the environment constant throughout validation meant each dashboard was evaluated under identical conditions — so the results genuinely reflect the reliability of the finished solution.

5. Testing Methodology

A structured, seven-step methodology was used to evaluate the functionality, accuracy, and reliability of the dashboard suite ahead of final submission. Each step was carried out using the finalized hospital dataset so that results stayed grounded in real, validated data.

Step 1 — Dataset Verification

The processed hospital dataset was reviewed to confirm it had been imported correctly into Power BI. Table relationships were checked, missing values were reviewed, and department names, patient records, and operational metrics were confirmed against the prepared dataset.

Step 2 — KPI Validation

Every healthcare KPI built with DAX was checked against the processed dataset, comparing the values shown on the dashboard with the expected values under different filter conditions.

Step 3 — Dashboard Functionality Testing

Each dashboard page was tested individually to confirm that KPI cards, charts, tables, and other visual elements displayed the correct information and responded properly to interaction.

Step 4 — Filter and Interaction Testing

Slicers, report filters, and cross-filtering behaviour were tested to confirm that user selections updated the right visuals dynamically, across different filter combinations.

Step 5 — Data Consistency Verification

Chart and KPI-card values were compared against the processed dataset — department-wise analysis, monthly trends, patient statistics, resource metrics, and operational indicators were all checked for accuracy.

Step 6 — Performance and Usability Testing

Loading time, visual responsiveness, layout consistency, chart readability, and alignment were assessed to confirm the reports stayed clear and suitable for analytical decision-making.

Step 7 — Result Documentation

Every validation activity was recorded with its objective, expected result, actual observation, and final status — the evidence base for this report.

Following this structured approach meant both the technical implementation and the analytical outputs of the dashboard were thoroughly verified — giving confidence that the solution is genuinely suitable for healthcare operations monitoring and analytical reporting.

6. Functional Testing

Functional testing verified that each dashboard component behaved according to the intended business requirements — covering KPI calculations, visual accuracy, interactive features, and filter behaviour across every dashboard page. Each test case ran against the finalized hospital dataset, with observed results checked against expected system behaviour.

Test cases were organized by the four dashboards developed for the project, so each dashboard could be assessed on its own merits while also confirming that the integrated suite worked correctly as a whole.

6.1 Executive Hospital Performance Dashboard

This dashboard gives a high-level overview of hospital operations — key performance indicators, operational trends, revenue analysis, occupancy monitoring, and departmental performance. Testing focused on the accuracy of KPI calculations, trend visualizations, departmental summaries, and filter responsiveness.

Functional Test Cases

Test ID	Test Scenario	Expected Result	Actual Result	Status
Test-01	Total Patients KPI	Correct count of unique patients	Matched the processed dataset	Pass
Test -02	Total Admissions KPI	Correct number of admissions	Matched the processed dataset	Pass
Test -03	Total Revenue KPI	Correct total revenue generated	Displayed correctly	Pass
Test -04	Average Length of Stay (LOS) KPI	Correct calculated average stay	Matched the implemented DAX logic	Pass
Test -05	Bed Utilization Rate KPI	Correct utilization percentage	Updated correctly with the dataset	Pass
Test -06	Occupancy Rate KPI	Correct occupancy percentage	Calculated correctly	Pass
Test -07	Monthly Admissions Trend	Values shown in chronological order	Accurate monthly data displayed	Pass
Test -08	Monthly Revenue Trend	Correct monthly revenue values	Matched the dataset	Pass
Test -09	Monthly Readmission Trend	Values update correctly by month	Displayed accurately	Pass
Test -10	Appointment Status Distribution	Correct category counts	Matched processed records	Pass
Test -11	Revenue by Department	Correct aggregation per department	Displayed correctly	Pass
Test -12	Average LOS by Department	Correct department-wise averages	Displayed accurately	Pass
Test -13	Monthly Occupancy Trend	Reflects monthly operational data	Displayed correctly	Pass
Test -14	Department Slicer	Updates all relevant visuals	Responded correctly to selection	Pass
Test -15	Gender Slicer	Filters patient-related visuals	Updated as expected	Pass

6.2 Patient Flow Analytics Dashboard

This dashboard tracks patient movement across the hospital journey. Validation covered admission and discharge statistics, patient demographics, readmission analysis, department-wise patient distribution, and interactive filtering.

Functional Test Cases

Test ID	Test Scenario	Expected Result	Actual Result	Status
Test -16	Total Patients KPI	Correct number of unique patients	Matched the processed dataset	Pass
Test -17	Total Admissions KPI	Correct number of admissions	Matched the processed dataset	Pass
Test -18	Total Discharges KPI	Correct number of discharged patients	Displayed correctly	Pass
Test -19	Average Length of Stay KPI	Correct calculated average stay	Matched the implemented DAX logic	Pass
Test -20	Total Readmissions KPI	Correct number of readmitted patients	Matched the dataset	Pass
Test -21	Readmission Rate KPI	Correct readmission percentage	Verified successfully	Pass
Test -22	Admissions by Department Chart	Correct department-wise admissions	Matched the processed dataset	Pass
Test -23	Discharges by Department Chart	Correct department-wise discharges	Accurate discharge counts	Pass
Test -24	Admissions vs Discharges Comparison	Accurate values for each department	Correct values displayed	Pass
Test -25	Gender Distribution Chart	Correct male / female distribution	Matched the processed dataset	Pass
Test -26	Readmissions by Department Chart	Correct aggregation per department	Displayed accurately	Pass
Test -27	Age Group Distribution Chart	Patients grouped into correct age bands	Matched the dataset	Pass

Testing Outcome

All KPI cards, demographic visualizations, and patient-movement analyses performed as expected. The dashboard accurately represented admissions, discharges, readmissions, and patient demographics, and responded correctly to interactive filters — no discrepancies were found against the processed healthcare dataset.

6.3 Department Analytics Dashboard

This dashboard validates departmental performance, workforce distribution, equipment allocation, and operational efficiency across hospital departments — including department-level KPIs, staffing information,

patient volume, equipment availability, and efficiency calculations, alongside interactive filtering and comparative analysis.

Functional Test Cases

Test ID	Test Scenario	Expected Result	Actual Result	Status
Test -28	Total Departments KPI	Correct number of departments	Matched the processed dataset	Pass
Test -29	Total Doctors KPI	Correct number of doctors	Verified successfully	Pass
Test -30	Total Nurses KPI	Correct number of nurses	Matched the processed dataset	Pass
Test -31	Total Equipment KPI	Correct number of equipment records	Displayed accurately	Pass
Test -32	Total Equipment Usage KPI	Correct equipment usage value	Matched the dataset	Pass
Test -33	Department Efficiency Score	Correct score per department	DAX calculation verified	Pass
Test -34	Doctors by Department Chart	Correct doctor count per department	Matched the dataset	Pass
Test -35	Nurses by Department Chart	Correct nurse count per department	Accurate staffing information	Pass
Test -36	Equipment Count by Department	Correctly grouped per department	Verified successfully	Pass
Test -37	Patient Volume by Department	Correct patient volume per department	Matched the processed dataset	Pass
Test -38	Rooms by Department Chart	Correct room count per department	Verified successfully	Pass

Testing Outcome

The Department Analytics Dashboard consistently displayed accurate departmental statistics and operational metrics. Staffing information, equipment allocation, patient distribution, and efficiency calculations all matched the processed dataset, and interactive filtering and comparison visuals responded correctly throughout.

6.4 Resource Utilization Dashboard

This dashboard verifies the effective utilization and availability of hospital resources — beds, wards, rooms, equipment, and staff allocation — with particular attention to equipment availability, bed occupancy, maintenance status, and monthly equipment-usage trends.

Functional Test Cases

Test ID	Test Scenario	Expected Result	Actual Result	Status
Test -39	Total Beds KPI	Correct total number of beds	Matched the processed dataset	Pass
Test -40	Total Wards KPI	Correct number of wards	Verified successfully	Pass
Test -41	Total Rooms KPI	Correct number of rooms	Matched the processed dataset	Pass
Test -42	Bed Utilization Rate KPI	Correct utilization percentage	Verified successfully	Pass
Test -43	Average Equipment Usage KPI	Correct average usage hours	Matched the processed dataset	Pass
Test -44	Bed Occupancy by Ward Chart	Accurate occupied-bed counts	Ward-wise occupancy matched	Pass
Test -45	Equipment Type Distribution	Correctly grouped by equipment type	Displayed correctly	Pass
Test -46	Staff Allocation by Department	Correct allocation per department	Verified successfully	Pass
Test -47	Monthly Equipment Usage Trend	Correct chronological order of hours	Matched the processed dataset	Pass

Testing Outcome

The Resource Utilization Dashboard performed as expected throughout validation. All resource-related KPIs, utilization metrics, and operational charts accurately reflected the dataset, filters updated visuals correctly, and no inconsistencies were found in equipment utilization, ward occupancy, staff allocation, or resource availability.

6.5 Dashboard Integration & Overall Validation

With each dashboard validated individually, the integrated solution was tested as a single analytical application — checking common filters, navigation, report responsiveness, and consistency of outputs across the whole suite.

Integration Test Cases

Test ID	Test Scenario	Expected Result	Actual Result	Status
Test -48	Dashboard Navigation	Navigate between all four dashboards without errors	Functioned correctly across all pages	Pass

Test ID	Test Scenario	Expected Result	Actual Result	Status
Test -49	Filter Consistency	Common filters update all applicable visuals / KPIs	Responded accurately to selections	Pass
Test -50	Overall Dashboard Performance	Loads, renders, and stays responsive	Performed smoothly, no issues	Pass

Overall Functional Testing Summary

In total, 50 functional test cases were executed across the four dashboards, covering KPI validation, visualization accuracy, interactive filtering, departmental analysis, patient flow monitoring, resource utilization, and dashboard integration.

Every test case passed, with no critical functional issues found. KPI calculations produced accurate results, visualizations stayed consistent with the underlying data, and interactive components responded correctly to user input — the dashboard suite meets the project's functional requirements and is ready for final documentation and submission.

7. KPI Validation

7.1 Overview

The dashboard suite is built on a set of healthcare KPIs that give meaningful insight into hospital operations, patient management, departmental performance, and resource utilization. Since these KPIs are the foundation everything else sits on, a dedicated validation pass confirmed that every calculated value accurately represented the processed dataset.

Validation focused on confirming that each KPI produced consistent results under different filter conditions while maintaining data integrity across dashboard pages. Every KPI was checked against the dataset and its corresponding DAX calculation to confirm it reflected the intended business logic — carried out once data cleaning, transformation, and KPI engineering were complete, to minimize inconsistencies before deployment.

Beyond raw numerical accuracy, validation also confirmed that KPI values updated dynamically as users interacted with dashboard filters — so the analytical insight stays reliable regardless of which department, patient category, or reporting period is selected.

7.2 KPI Validation Results

KPI	Validation Method	Expected Result	Actual Result	Status
Total Patients	Compared with processed patient records	Correct count of unique patients	Matched dataset	Pass

KPI	Validation Method	Expected Result	Actual Result	Status
Total Admissions	Compared with admission records	Correct total admissions	Matched dataset	Pass
Total Revenue	Verified against revenue calculations	Correct total revenue	Calculated correctly	Pass
Average LOS	Compared with implemented DAX measure	Correct average patient stay	Verified successfully	Pass
Occupancy Rate	Verified using occupancy calculation	Correct occupancy percentage	Matched calculated value	Pass
Readmission Rate	Compared with processed admission data	Correct readmission percentage	Validated successfully	Pass
Total Readmissions	Compared with patient readmission records	Correct readmission count	Matched dataset	Pass
Bed Utilization Rate	Verified against available bed records	Correct utilization percentage	Verified successfully	Pass
Department Efficiency Score	Compared with implemented DAX calculation	Correct efficiency score per department	Validated successfully	Pass
Equipment Availability	Compared with equipment inventory records	Correct availability information	Displayed accurately	Pass
Average Equipment Usage	Compared with equipment usage records	Correct average usage hours	Verified successfully	Pass

7.3 KPI Validation Outcome

Every healthcare KPI in the dashboard passed validation. Calculated values stayed consistent with the processed dataset and responded accurately to interactive filters without compromising the integrity of the results — each KPI reflected its intended business logic and delivered information hospital teams can rely on for operational monitoring and decision-making.

DAX measures performed correctly across every dashboard page, keeping KPI cards, supporting visualizations, and underlying healthcare records in sync — no calculation discrepancies or data inconsistencies were observed.

Based on these results, every implemented KPI met the analytical requirements of the project and was confirmed suitable for the final dashboard solution.

8. Dashboard Performance & User Interaction Testing

8.1 Overview

Beyond functional requirements and KPI accuracy, the dashboard was evaluated for overall performance and user interaction — confirming it stays responsive, visually consistent, and easy to use under real interaction patterns, without sacrificing data accuracy or user experience.

This phase looked at loading behaviour, visual responsiveness, interaction between report components, filter execution speed, and overall usability, running multiple interaction scenarios to confirm dashboard elements responded correctly while keeping analytical results consistent.

8.2 User Interaction Testing

The dashboard's interactive capabilities were tested to confirm users can explore hospital data efficiently through filters and visual interactions — applying different combinations of slicers and checking that the corresponding visuals updated correctly and consistently.

ID	Interaction Tested	Expected Result	Actual Result	Status
UI-01	Department Slicer	Updates all relevant department visuals	Visuals updated correctly	Pass
UI-02	Gender Slicer	Filters patient-related KPIs and charts	Responded correctly	Pass
UI-03	Equipment Type Slicer	Updates equipment-related visuals	Correct information displayed	Pass
UI-04	Ward Filter	Updates ward and bed utilization visuals	Filter operated correctly	Pass
UI-05	Multiple Filter Selection	Applies several filters without conflicts	Maintained accurate results	Pass
UI-06	KPI Synchronization	KPI cards update with applied filters	Values refreshed correctly	Pass
UI-07	Chart Cross-Filtering	Selections update related charts	Cross-filtering worked correctly	Pass
UI-08	Filter Reset	Removing filters restores default state	Returned to default values	Pass

8.3 Dashboard Performance Validation

Performance testing evaluated the dashboard's responsiveness and stability under normal use, reviewing each page across different interaction scenarios to confirm visualizations loaded correctly and performance stayed within acceptable limits.

ID	Performance Check	Expected Result	Actual Result	Status
PF-01	Dashboard Loading	Loads successfully	Loaded without errors	Pass
PF-02	KPI Rendering	KPI cards display correctly	All cards rendered correctly	Pass
PF-03	Chart Rendering	Charts display without visual issues	Visuals rendered successfully	Pass
PF-04	Filter Response Time	Filters update within acceptable time	Response time satisfactory	Pass
PF-05	Dashboard Navigation	Navigation between pages works correctly	Navigation performed successfully	Pass
PF-06	Visual Alignment	Layout remains properly aligned	No alignment issues observed	Pass
PF-07	Data Consistency	Visuals remain synchronized after interaction	Data remained consistent	Pass
PF-08	Overall Dashboard Stability	Remains stable throughout testing	No unexpected behaviour observed	Pass

8.4 Performance Testing Outcome

The dashboard held up well throughout testing. Interactive filters responded consistently, KPI cards refreshed accurately, and visualizations stayed synchronized across different user interactions — no significant performance degradation, rendering issues, or functional inconsistencies were observed.

The evaluation also confirmed a clear, consistent layout across every report page, giving users a reliable and intuitive analytical experience overall — the dashboard met the expected standards for usability, responsiveness, and operational performance.

9. Testing Summary

9.1 Overall Testing Summary

A comprehensive testing process evaluated the functionality, accuracy, performance, and usability of the Hospital Operations & Patient Analytics Dashboard, covering all four dashboards: Executive Hospital Performance, Patient Flow Analytics, Department Analytics, and Resource Utilization.

The validation process combined functional testing, KPI verification, dashboard interaction testing, filter validation, and performance evaluation. Every component was checked against the processed hospital dataset to confirm results stayed accurate and consistent under real user interaction.

In total, 50 functional test cases were executed, covering KPI cards, charts, visualizations, filters, DAX calculations, dashboard responsiveness, and integrated report functionality — the results confirm the dashboard behaves according to the expected business requirements and produces reliable analytical output.

9.2 Test Execution Summary

77 Total Tests Performed	77 Tests Passed	100% Success Rate
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Category	Result
Total Dashboards Tested	4
Total Functional Test Cases Executed	50
KPI Validation Tests	11
Dashboard Interaction Tests	8
Performance Validation Tests	8
Total Tests Performed	77
Test Cases Passed	77
Test Cases Failed	0
Success Rate	100%

9.3 Key Observations

- All KPI calculations produced values consistent with the processed hospital dataset.
- Interactive filters updated the relevant dashboard components correctly without affecting data integrity.
- Department-wise, patient-wise, and resource-related visualizations displayed accurate analytical information.
- DAX measures performed consistently across all dashboard pages.
- Dashboard navigation and report responsiveness remained stable throughout testing.
- Visual layouts, chart formatting, labels, and KPI cards maintained a consistent, user-friendly presentation.
- No critical functional defects, calculation errors, or visualization inconsistencies were identified during validation.
- The dashboard provided accurate operational insight that supports healthcare monitoring and decision-making.

9.4 Conclusion

Testing and validation confirmed that the Hospital Operations & Patient Analytics Dashboard satisfies the functional and analytical requirements set for the project. Every dashboard was successfully validated for KPI accuracy, visualization correctness, interactive behaviour, and overall performance.

The dashboard held up reliably across a wide range of test scenarios — KPI verification, department-level analysis, patient flow monitoring, resource utilization tracking, and interactive filtering — maintaining consistent analytical results and responding appropriately to user interaction without compromising accuracy or usability.

On the strength of complete functional testing, KPI validation, interaction testing, and performance evaluation, the dashboard is ready for final documentation, project submission, and demonstration — a dependable Business Intelligence platform for monitoring hospital operations, supporting analytical reporting, and enabling informed decision-making within the scope of this healthcare analytics project.